

PREDICTING STUDENT PRODUCTIVITY USING STUDY HABITS AND LIFESTYLE FACTORS

Author: Zheng Chao, Jiajun Li, Kunji Yang

Background

Student productivity can be influenced by study habits, digital behavior, wellbeing, and academic engagement. We first used linear regression as a baseline, then applied Random Forest Regression to capture more complex patterns and identify key predictors. Other poorly performing models are not discussed here.

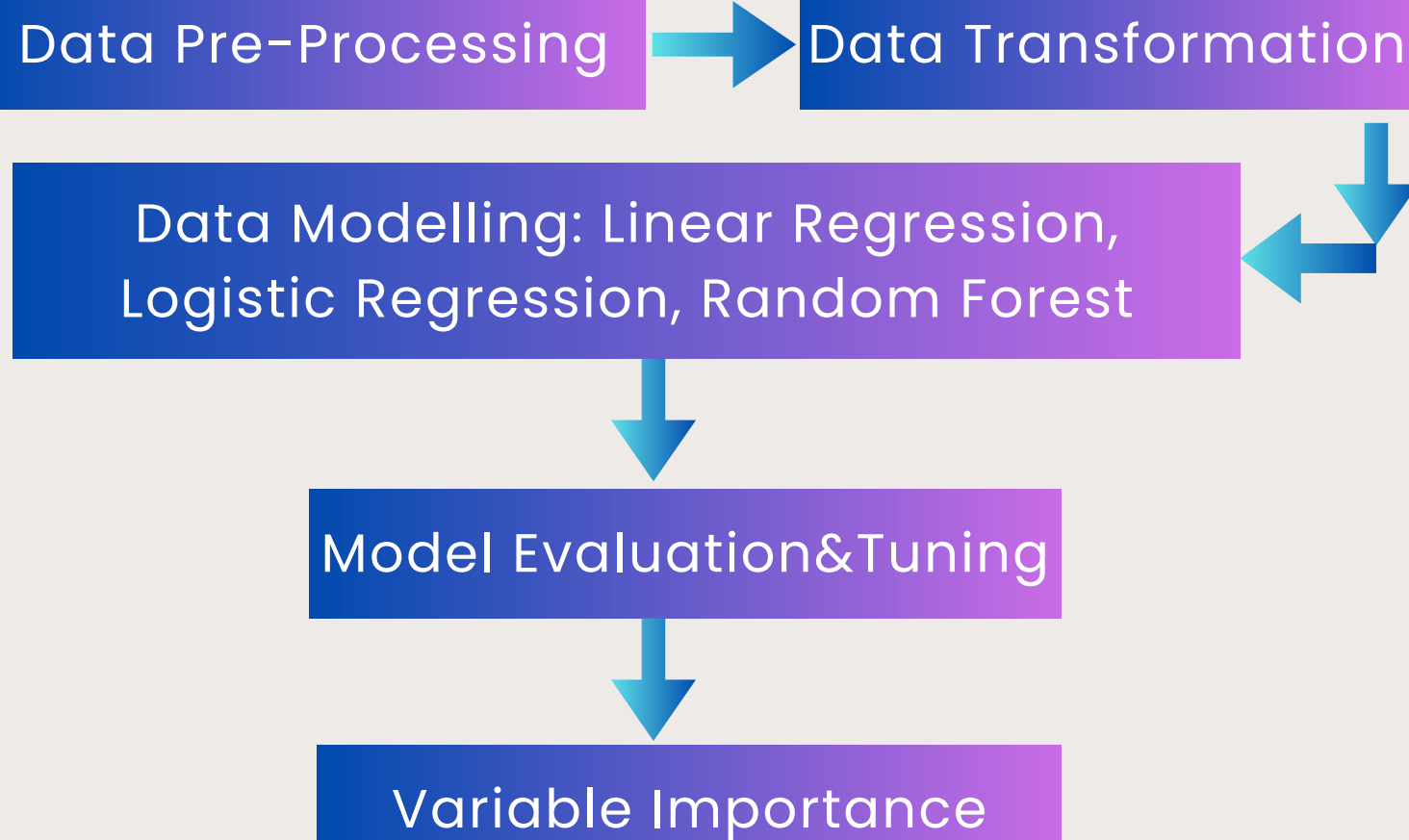
Objective

This project aims to predict student productivity from behavioural and lifestyle factors. We explored linear regression, logistic regression, K-means clustering, and Random Forest Regression. Random Forest Regression was selected as the main model because it directly predicts the continuous productivity score and achieved the best performance.

Dataset

The dataset contains student-level information about study habits, lifestyle behaviours, academic engagement, and productivity. The target variable is productivity_score, which is treated as a continuous outcome for regression analysis.

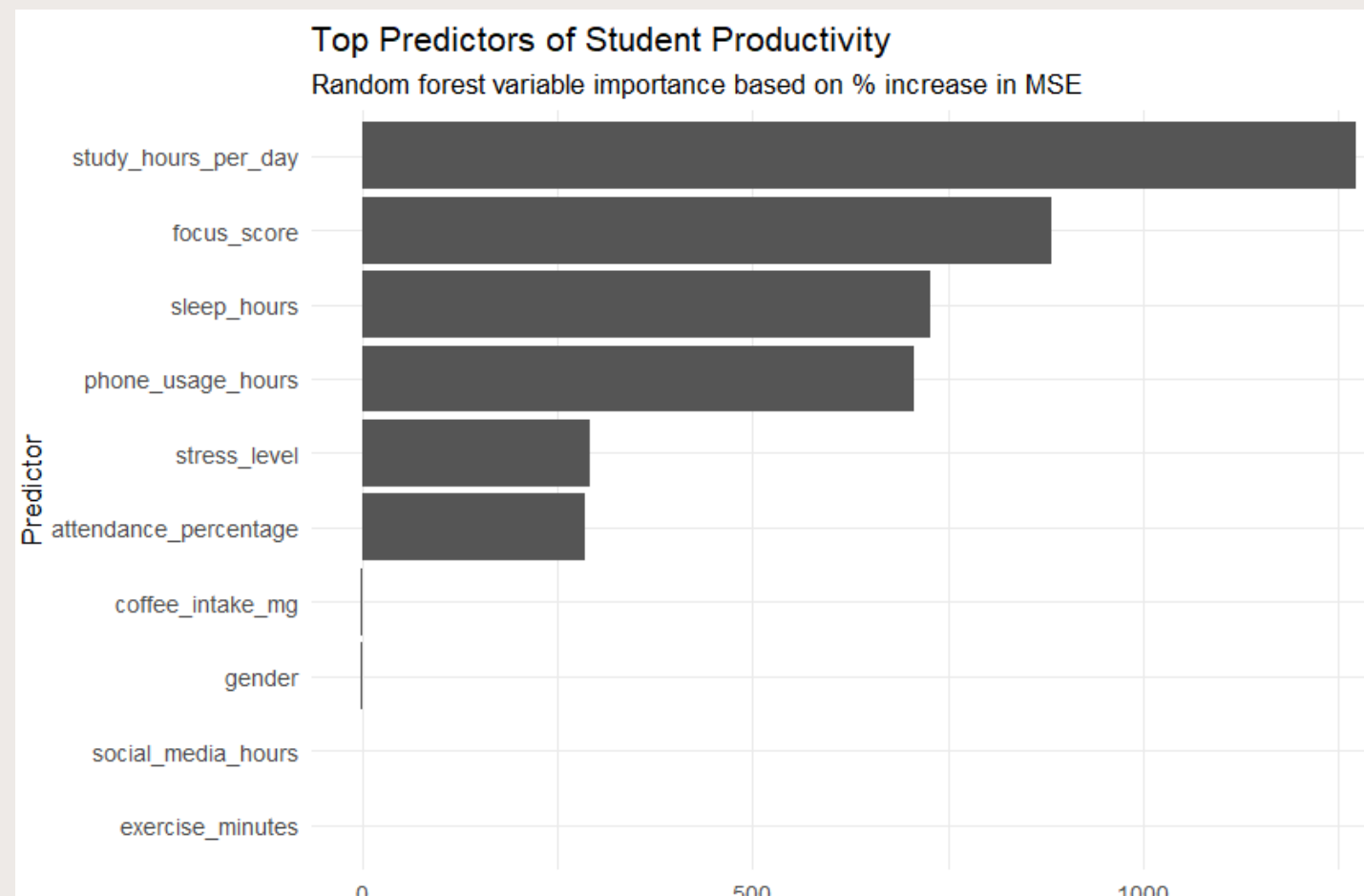
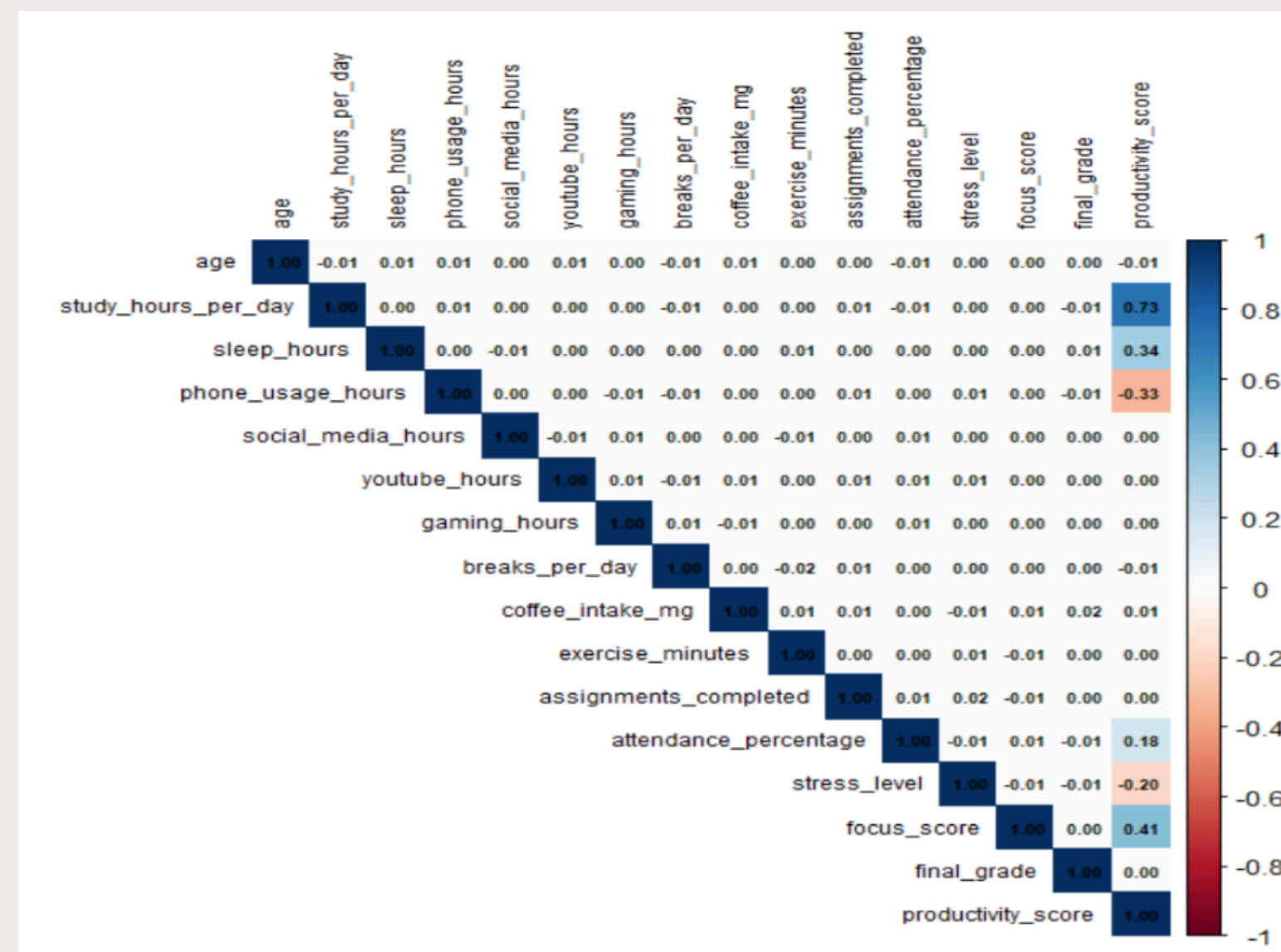
Methodology



Result

Model	R ²	RMSE
Linear Regression	0.884	5.359
Random Forest	0.979	2.482

Random Forest Regression achieved the best performance, with a higher R² and lower RMSE than Linear Regression.



Limitations & Future Work

- Weak relationships between variables limited model performance.
- Some important factors, such as motivation and learning environment, were not captured.
- Future work could collect stronger predictors and test more advanced models.